

LLM Prompts

We structure the prompts for LLM-based analysis of explanations similar to Rathje et al. (2024), who gave models minimal context when rating LIWC categories. Prompts for past/future language, perceptual details, causal/analytical language, comparative language, and cleverness are shown in Figures 5, 6, 7, 8, and 9 respectively. Our prompt for Study 2 is listed in Figure 10.

Additional Correlation and Distribution Results

Figure 11 summarizes differences in facet correlations for CLAUDE-3.5-HAIKU. Trends are similar to that of GPT-4O-MINI, though magnitudes in correlation are not as strong in many cases. Figures 12 and 13 report Kappa agreement among the facets for GPT-4O-MINI and CLAUDE-3.5-HAIKU respectively. Trends are largely similar to the Pearson correlations: agreement still rises sharply among all pairs in the example condition (pointing to homogenization) and cleverness is still the weakest predictor of originality for GPT-4O-MINI (though the opposite is true for CLAUDE-3.5-HAIKU). Rating distributions for cleverness, remoteness, and uncommonness for humans compared against GPT-4O-MINI are reported in Figure 14.

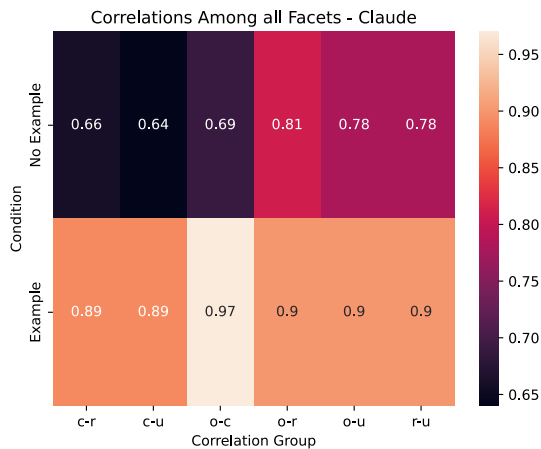


Figure 11: Pearson correlations among pairwise Likert ratings for CLAUDE-3.5-HAIKU in both conditions. o = originality, c = cleverness, u = uncommonness, r = remoteness.

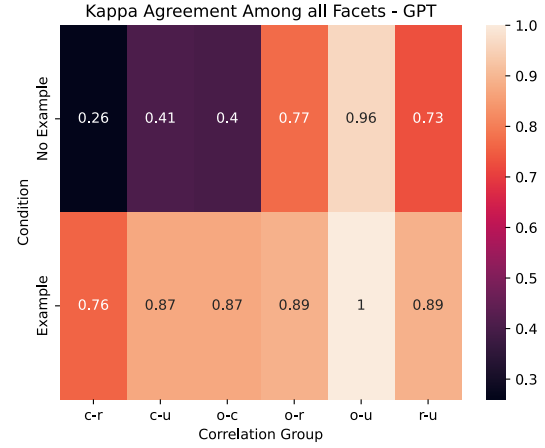


Figure 12: Kappa agreement statistics GPT-4O-MINI in both conditions. o = originality, c = cleverness, u = uncommonness, r = remoteness.

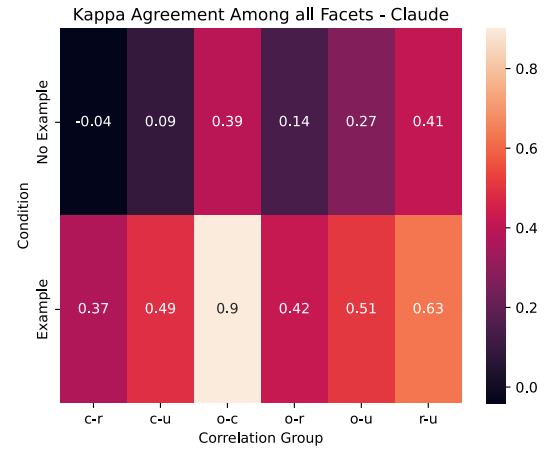


Figure 13: Kappa agreement statistics CLAUDE-3.5-HAIKU in both conditions. o = originality, c = cleverness, u = uncommonness, r = remoteness.

Additional Explanation Results

Results from CLAUDE-3.5-HAIKU, and from CLAUDE-3.5-SONNET for both human and GPT-4O-MINI explanations, are presented in Figure 15. For CLAUDE-3.5-HAIKU, both CLAUDE-3.5-SONNET and GPT-4O found a significant difference in past/future language use (claude U = 105253.5, $p < 0.001$; GPT U = 105048.5, $p < 0.001$), comparative language (claude U = 88600, $p < 0.05$; GPT U = 80501.5, $p < 0.001$), and cleverness (claude U = 103821, $p < 0.001$; GPT U = 99598.5, $p < 0.01$), but found no significant difference for perceptual details as a function of condition.

Are the words in this text past-focused or future-focused? ANSWER ONLY WITH A NUMBER: 1 if past-focused, 2 if future-focused. Here is the text: [RESPONSE]

Figure 5: Past/future language prompt. [RESPONSE] is filled with the participant explanation.

Perceptual details refer to the process of perceiving and are indicated by words like observing, seen, heard, feeling, listen, or touch, among others. Does this text contain perceptual details? ANSWER ONLY WITH A NUMBER: 1 if the text doesn't contain perceptual details, 2 if the text does contain perceptual details. Here is the text: [RESPONSE]

Figure 6: Perceptual details prompt. [RESPONSE] is filled with the participant explanation.

Casual/analytical markers refer to performing structured evaluation of ideas. The cause/insight/analytical LIWC category words are example markers of causal processes. Casual markers are contrasted with intuitive markers, which do not indicate a structured evaluation and instead mark an intuitive (e.g., gut feeling) one. Does this text contain causal/analytical markers? ANSWER ONLY WITH A NUMBER: 1 if the text uses an intuitive and not analytical approach, 2 if the text uses a structured and analytical approach. Here is the text: [RESPONSE]

Figure 7: Causal/analytical prompt. [RESPONSE] is filled with the participant explanation.

Comparative language tracks explicit references to standards or examples and includes comparing or contrasting ideas or drawing distinctions among ideas. Does this text contain comparative language? ANSWER ONLY WITH A NUMBER: 1 if the text doesn't contain comparative language, 2 if the text does contain comparative language. Here is the text: [RESPONSE]

Figure 8: Comparative language prompt. [RESPONSE] is filled with the participant explanation.

Does this text refer to the cleverness, wittiness, shrewdness, or ingenuity of an idea? ANSWER ONLY WITH A NUMBER: 1 if the text doesn't refer to cleverness, 2 if the text does refer to cleverness. Here is the text: [RESPONSE]

Figure 9: Cleverness prompt. [RESPONSE] is filled with the participant explanation.

You are evaluating the originality of solutions to this engineering design problem: [PROBLEM]

You will rate solutions on multiple criteria using 1-5 scales and provide explanations. Your response must follow this exact format:

ORIGINALITY: [1-5 or NA]
UNCOMMON: [1-5]
REMOTE: [1-5]
CLEVER: [1-5]
EXPLANATION: [1-2 sentences explaining the originality rating from a STEM perspective]

Rating scales:

Originality:

- 1: Very Unoriginal
- 2: Unoriginal
- 3: Neutral
- 4: Original
- 5: Very Original
- NA: For solutions that are too short or unclear

Uncommon:

- 1: Very common
- 2: Common
- 3: Neutral
- 4: Uncommon
- 5: Very uncommon

Remote:

- 1: Very unremote
- 2: Unremote
- 3: Neutral
- 4: Remote
- 5: Very remote

Clever:

- 1: Very unclever
- 2: Unclever
- 3: Neutral
- 4: Clever
- 5: Very clever

Evaluation criteria:

- Uncommon: Consider if the solution is rare. Common solutions given by many people score low. A unique solution may score high, unless it's just strange rather than original.
- Remote: Consider how far the solution is from everyday ideas. Non-obvious solutions score higher. Solutions similar to common approaches score lower.
- Clever: Consider the insight and wit shown. Clear, insightful solutions score higher. Unclear solutions score lower. Even common solutions can score high if presented cleverly.

Provide your STEM-based explanation focusing solely on the originality rating. Be specific about why the solution is or isn't original from an engineering perspective.

Figure 10: Prompt for LLM data collection in Study 2. [PROBLEM] is filled with the design problem.

Explanation Qualitative Analysis

We include examples of human and GPT-4O-MINI explanations for responses from the problem “assist people with learning impairments retain information” in Table 1. We find that human explanations tend to be shorter, more egocentric (e.g., “I think”), and more variable in their structure and areas of focus. LLM explanations are more formulaic and make more direct references to prior ideas. We also find that LLMs sometimes address cleverness, remoteness, or uncommonness in their explanations, though in our prompt we emphasize that explanations should focus on originality, which supports the idea that LLMs homogenize these facets with originality.

Table 1: Sample human (first three) and GPT-4O-MINI (last three) explanations for the same prompt.

Explanations
I think this idea implements gamification and behavioral conditioning in an interesting way. I'm not sure how adaptive it would be for learning impairments which can be varied.
This is original because it mimics the human encoding process for information. This will also help many people with disabilities or speech impairments, as they can listen to recordings as many times as needed.
Said and done. We would need new and improved writing tools to assist, but now that we have CGPT-4 and all these included advancements within said tech, it's trying to find what else can work because right now there really isn't anything new and improved, unless it's expensive and most people haven't heard of it.
This solution is original as it combines a common tool (flashcards) with a unique application (phone lock screen) to create a novel way to reinforce learning. The integration of learning into a daily-use device is a clever approach, though the concept of using a lock screen for educational purposes is somewhat familiar, which affects its remoteness.
This solution is moderately original as it builds on existing educational technology concepts but introduces a specific focus on time management and personalized learning. While the idea of using apps for learning is common, the clever integration of gauging time spent on topics adds a thoughtful layer to the approach.
The idea of using art as a tool for learning is not particularly original, as it has been explored in various educational contexts. While art can enhance memory retention, the concept itself is common and lacks innovative engineering applications or methods.

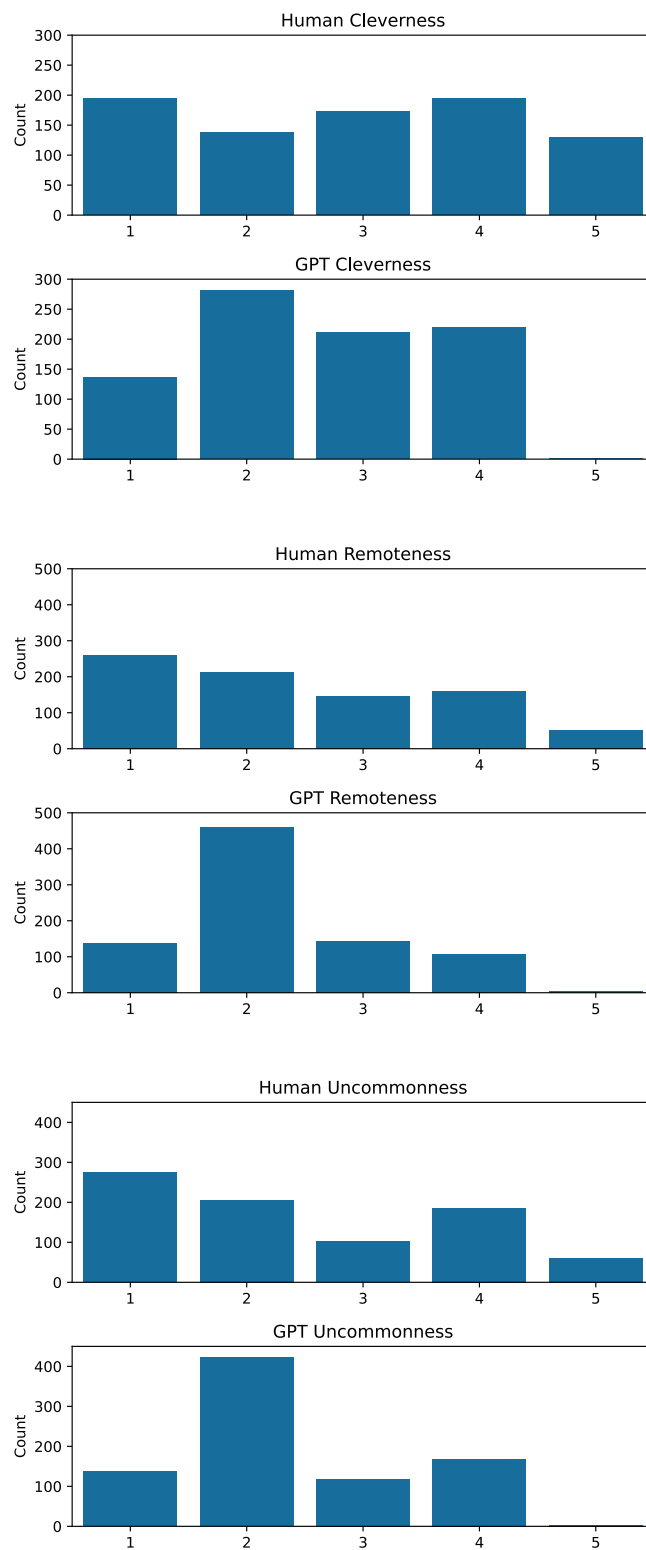


Figure 14: Rating distributions for cleverness, remoteness, and uncommonness for humans and GPT-4O-MINI.

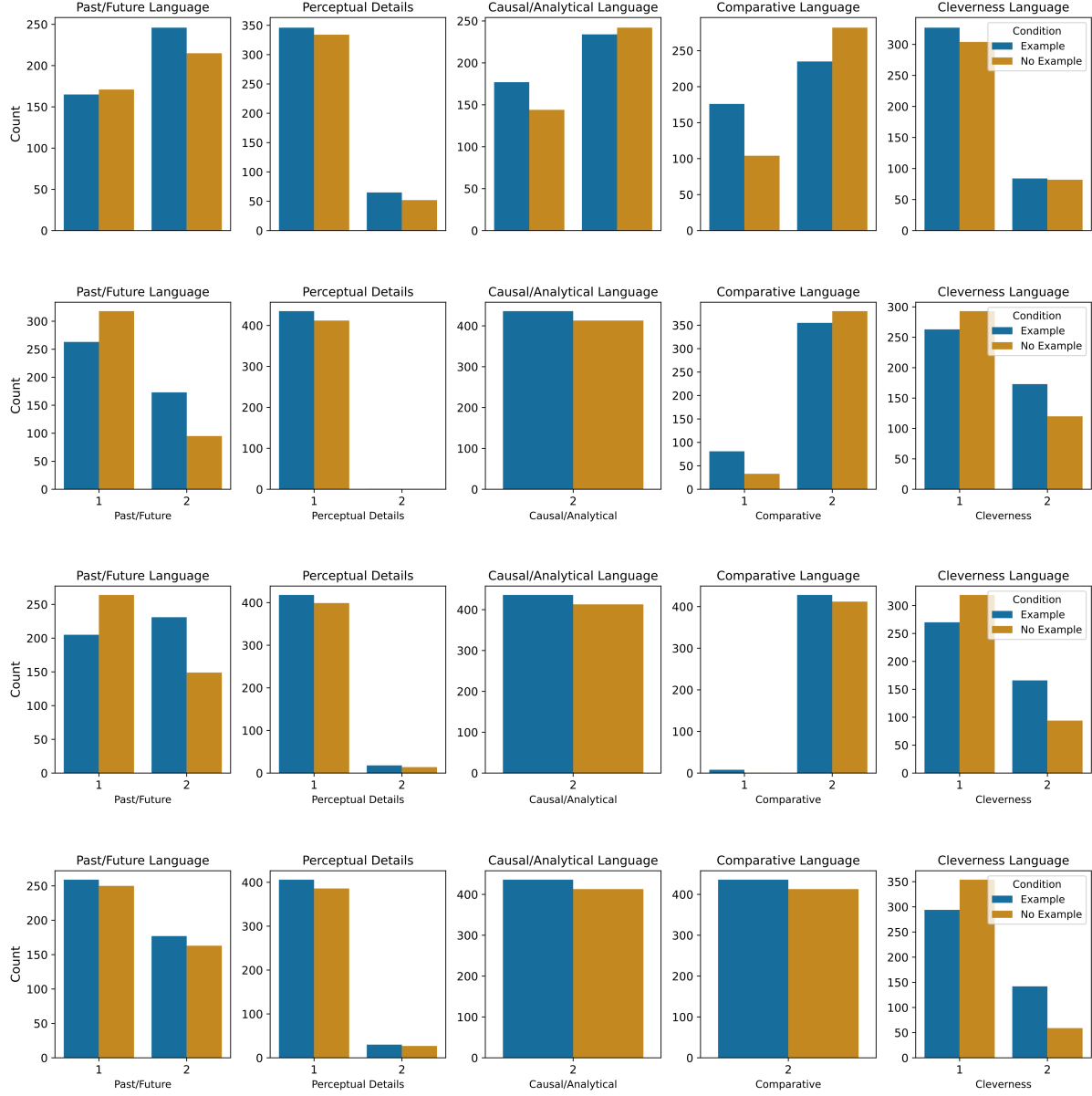


Figure 15: Comparisons between linguistic marker use for all populations and LLM raters not reported in the main paper. Row 1: human explanations rated by CLAUDE-3.5-SONNET. Row 2: GPT-4O-MINI explanations rated by CLAUDE-3.5-SONNET. Row 3: CLAUDE-3.5-HAIKU explanations rated by CLAUDE-3.5-SONNET. Row 4: CLAUDE-3.5-HAIKU explanations rated by GPT-4O.